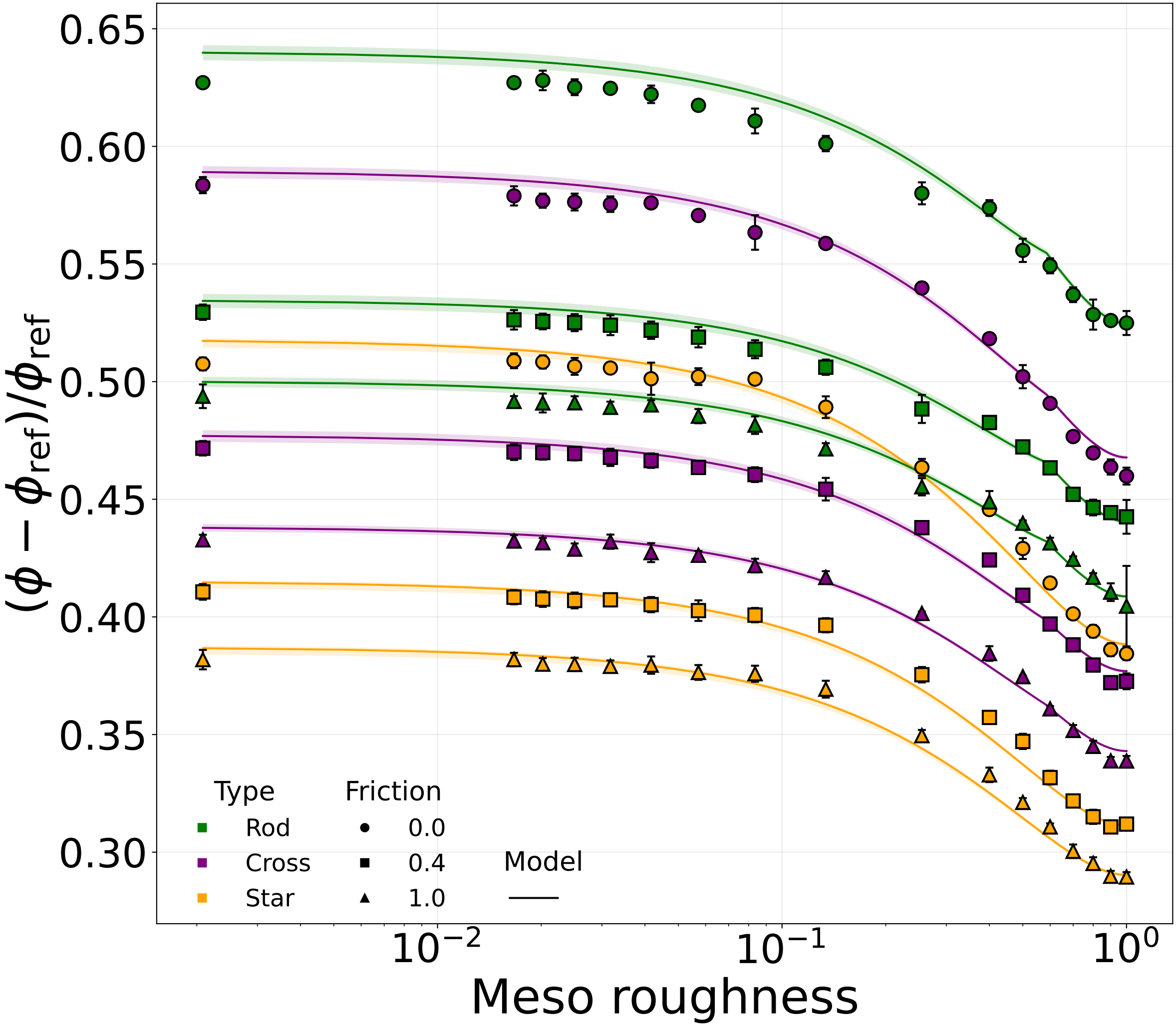

0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
$(\phi - \phi_{ref})/\phi_{ref}$
10^{-2}
10^{-1}
10^{0}
Meso roughness
Type
Rod
Cross
Star
Friction
0.0
0.4
1.0
Model